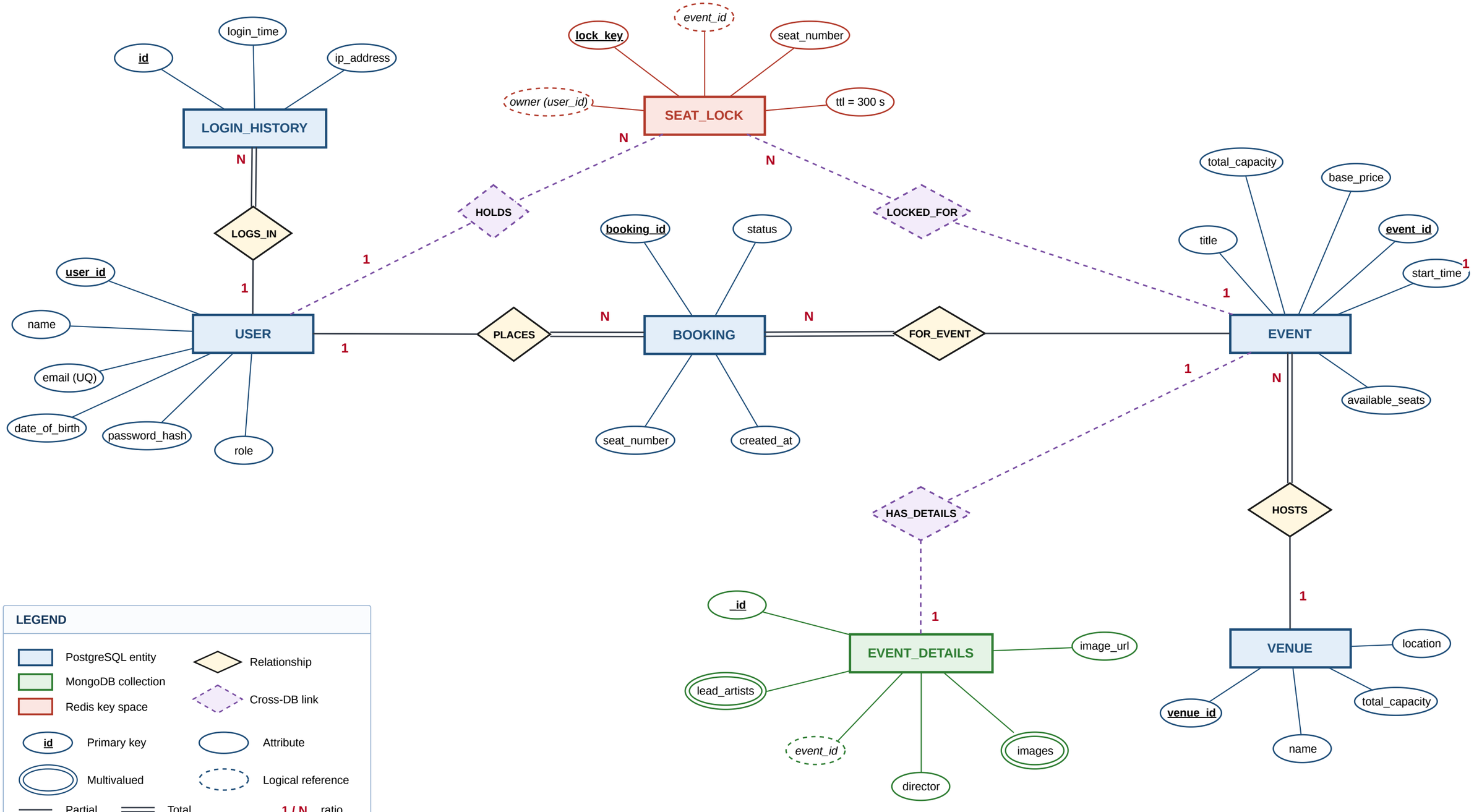
High-Concurrency Ticketing and Event Management Web Application

Entity-Relationship (ER) Diagram



LEGEND

- PostgreSQL entity
- MongoDB collection
- Redis key space
- Relationship
- Cross-DB link
- **id** Primary key
- Attribute
- Multivalued
- Logical reference
- Partial
- Total
- **1 / N** ratio

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Relational Schema Diagram

PostgreSQL

users

PK	**user_id**	UUID	
	name	VARCHAR(120)	NOT NULL
	email	VARCHAR(320)	UNIQUE
	password_hash	TEXT	NOT NULL
	role	VARCHAR(32)	NOT NULL
	date_of_birth	DATE	

PostgreSQL

bookings

PK	**booking_id**	UUID	
FK1	user_id	UUID	NOT NULL
FK2	event_id	UUID	NOT NULL
	seat_number	VARCHAR(20)	NOT NULL
	status	VARCHAR(16)	NOT NULL
	created_at	TIMESTAMPTZ	NOT NULL

PostgreSQL

events

PK	**event_id**	UUID	
FK	venue_id	UUID	NOT NULL
	title	VARCHAR(220)	NOT NULL
	start_time	TIMESTAMPTZ	NOT NULL
	base_price	NUMERIC(10,2)	NOT NULL
	total_capacity	INTEGER	NOT NULL
	available_seats	INTEGER	NOT NULL

PostgreSQL

venues

PK	**venue_id**	UUID	
	name	VARCHAR(180)	NOT NULL
	total_capacity	INTEGER	NOT NULL
	location	VARCHAR(220)	NOT NULL

PostgreSQL

login_history

PK	**id**	UUID	
FK	user_id	UUID	NOT NULL
	login_time	TIMESTAMP	NOT NULL
	ip_address	VARCHAR(45)	

Redis key

seat_lock

PK	**lock_key**	String	
FK1	event_id	(in key)	
	seat_number	(in key)	
FK2	user_id	(value)	
	TTL	300 seconds	

MongoDB collection

event_details

PK	**_id**	ObjectId	
FK	event_id	String	
	images	Array<String>	
	image_url	String	
	director	String	
	lead_artists	Array<String>	

LEGEND

- PostgreSQL Foreign Key (FK)
- Logical Cross-Database Reference
- 1 / N Ratio

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